

- 4 (a) A cell of e.m.f. $E = 3.0 \text{ V}$ and internal resistance $r = 0.50 \, \Omega$ is connected to three bulbs X, Y and Z as shown in the circuit of Fig. 4.1.

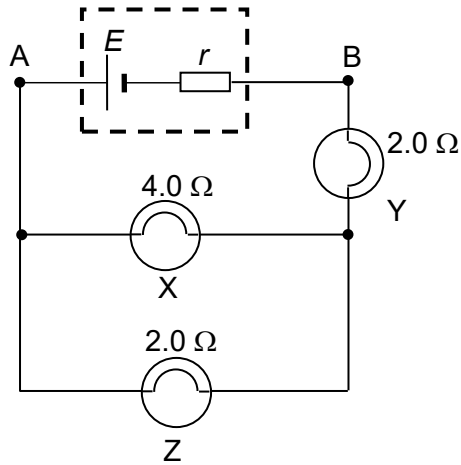


Fig. 4.1

- (i) List the light bulbs X, Y and Z in order of *increasing* brightness.

in order of increasing brightness: [1]

- (ii) Calculate the effective resistance of the three bulbs in the circuit of Fig. 4.1.

effective resistance = Ω [2]

- (iii) Calculate the potential difference across AB.

potential difference = V [2]

- (b) The circuit of Fig. 4.1 is connected to a potentiometer that is made up of a 6.0 V battery in series with a resistance $R = 1.0\ \Omega$ and a uniform wire CD of length 100.0 cm and resistance $2.0\ \Omega$ as shown in Fig. 4.2.

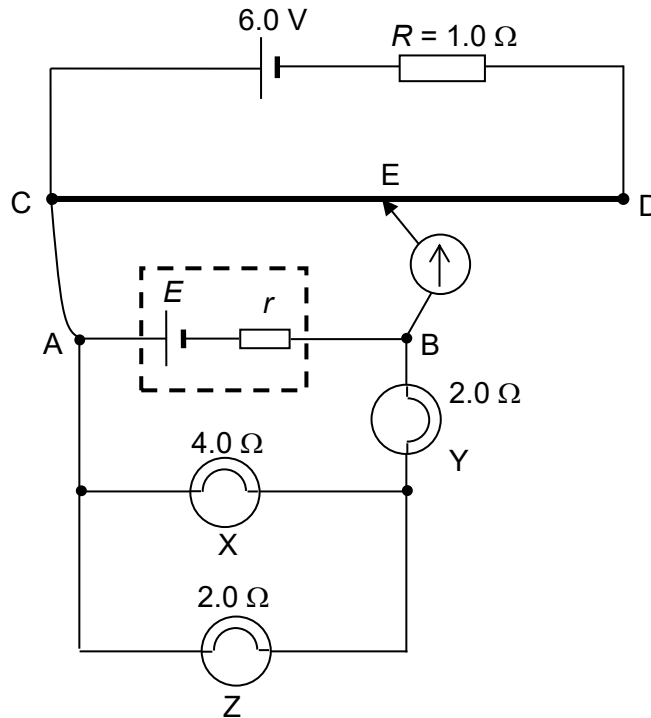


Fig. 4.2

- (i) Show that the potential difference across CD is 4.0 V when the galvanometer shows null deflection.

[1]

- (ii) Determine the balance length CE.

balance length CE =cm [2]

- (iii) Explain how, if at all, the balance length CE will be affected if bulb X is replaced by a bulb of lower resistance.

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..... [2]

[Total: 10]